USN. | | | | | | | | | | [24MTASE121]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY-2025

PYTHON PROGRAMMING FOR ENGINEERS

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

			CO's	Blooms Level	Marks
1.	a.	What is meant by selection, iteration and sequence control structures?	01	L2	10
	b.	Explain type conversion in Python with examples.	01	L2	10
2.	a.	Write a program to read the marks of three subjects and find the average of them.	02	L3	10
	b.	Write a short note on data types in Python.	02	L2	10
3.	a.	Explain tuples are immutable in Python programming with example.	03	L2	10
	b.	Explain looping control statements in Python with a syntax and example to each.	03	L2	10
		OR			
4.	a.	Explain append() and index() functions with respect to lists in Python.	03	L2	6
	b.	What are functions? Explain python function with parameters and return functions	03	L2	8
	c.	Develop a python program to find mean, variance and standard deviation, for a sequence of number data.	03	L3	6
5.	a.	What are packages? Give an example of package creation in Python	04	L2	10
	b.	Write a program to enter a number in Python and print its octal and hexadecimal equivalent.	04	L2	10
		OR			
6.	a.	Write a simple program which illustrates Handling Exceptions.	04	L2	6
	b.	List some few common exception types and explain when they occur.	04	L2	10
	c.	What is the difference between break and continue statement?	04	L2	4
7.	a.	What is Matplotlib? Write Python program to plot line chart by assuming your own data and explain the various attributes of line chart.	05	L3	10

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	b.	Synthesize the steps required to download data in CSV format from a web application built with Django	**05**	**L3**	**10**
		OR			
8.	**a.**	Outline the steps to create a basic web application using Django that displays a Matplotlib graph	**05**	**L3**	**10**
	b.	Explain Data Visualization and recognize its use. Write a Python code segment to visualize line chart and scatterplot with example	**05**	**L3**	**10**